MIVA OPEN UNIVERSITY — SCHOOL OF POSTGRADUATE STUDIES

Master of Information Technology (MIT) | MIT 8103: Advanced Database Systems

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Academic Session: 2026/2027 (First Semester CA) **Submission Platform:** LMS & GitHub

GitHub Repository: https://github.com/peace-ekoh/advanced-database-portfolio

1. EXECUTIVE SUMMARY & SELECTED ENTERPRISE CASE STUDY

NovaMart Retail Logistics & Storefront is an omnichannel e-commerce retail enterprise serving urban and regional consumers across West Africa. The enterprise distributes consumer electronics, personal computing, home appliances, fashion, and FMCG provisions. The database architecture is engineered to satisfy intense concurrent demand during regional promotional campaigns ("Black Friday"), supporting shoppers, warehouse inventory controllers, logistics dispatch operators, and automated payment gateways.

Rather than forcing all business workloads into a single database engine, NovaMart implements a **Polyglot Persistence Architecture**: (1) *MySQL 8.0 (InnoDB)* for ACID-compliant orders, financial payments, and stock audit ledgers; (2) *MongoDB (Document Store)* for dynamic, polymorphic product specifications and embedded customer reviews; and (3) *Redis (Key-Value)* for ephemeral shopping carts (with 7-day TTL expiration) and sub-millisecond stock decrement caching.

2. PORTFOLIO ACTIVITY 1: DATABASE DESIGN & RELATIONAL MODELLING (7 MARKS)

An 8-table relational schema was implemented in MySQL 8.0 InnoDB (`customers`, `customer_addresses`, `categories`, `products`, `orders`, `order_items`, `payments`, `inventory_logs`) with strict foreign keys (`ON DELETE CASCADE / RESTRICT`) and check constraints (`stock_quantity >= 0`, `unit_price > 0`).

Mathematical Normalisation Proof (1NF to 3NF/BCNF):

- **1NF:** All attribute values are atomic; repeating product collections were isolated into `order_items` and multi-valued delivery addresses into `customer_addresses`.
- **2NF:** In composite relation `order_items(order_id, product_id)`, non-prime attributes (product descriptions and current catalog prices) depend strictly on `product_id`. These were decoupled into `products`, ensuring every non-prime attribute in `order_items` depends entirely on the full composite key.
- **3NF / BCNF:** Eliminated transitive dependencies: `order_id → customer_id → customer_email` and `product_id → category_id → category_name`. Isolating categories and customers into independent relations ensures that for every functional dependency $X \rightarrow Y$, X is a superkey, completely eliminating update, insertion, and deletion anomalies.

3. PORTFOLIO ACTIVITY 2: QUERY PROCESSING & OPTIMISATION (7 MARKS)

The relational database was scaled with a benchmark workload of **9,010 Orders**, **9,015 Order Items**, and **9,010 Payments**. Three complex analytical queries were implemented and profiled using MySQL 8.0's `EXPLAIN ANALYZE` engine.

Analytical Business Query	Key Operations & SQL Clauses	Empirical Business Output
Category Revenue & Profit Margins	Multi-table `JOIN` (4 tables), `GROUP BY`, `SUM`, `FORMAT`, `ORDER BY`	Laptops: ₦4.38B Rev (13.16% margin); Phones: ₦2.46B Rev; Fashion: ₦262.5M Rev (29.09% margin).
High-Value Customer Profiling (LTV)	Conditional filtering, `DATEDIFF` recency, loyalty points ≥ 300, `HAVING`	Chinedu Okafor: 229 orders, ₦506.7M spent; Emeka Nwosu: 232 orders, ₦457.6M spent.
Inventory Velocity & Restock Priority	Window functions: `DENSE_RANK() OVER()`, `LEFT JOIN`, dynamic `CASE`	Categorized SKU sales velocity and identified items requiring warehouse replenishment.

> **Optimization Benchmark Results:** On the high-frequency order filtering query, introducing composite B-Tree index `idx_orders_status_date(order_status, order_date DESC)` **completely eliminated an expensive in-memory Filesort**, reduced rows scanned from **3,802 down to 1,280** (66.3% fewer reads), reduced query cost from **915.60 to 652.86**, and slashed latency from **25.47 ms to 14.74 ms (42.1% speedup)**. Covering index range scan completed in just **3.07 ms**.

4. PORTFOLIO ACTIVITY 3: TRANSACTIONS & CONCURRENCY CONTROL (7 MARKS)

ACID guarantees were demonstrated through functional transaction scripts and multi-threaded Python simulations:

- **Atomic Checkout (COMMIT):** Customer 2 purchased an iPhone 15 Pro (₦1.45M). Stock decremented (25 → 24), order 9388 created, and payment recorded atomically under a single commit.
- **Payment Failure & Full Abort (ROLLBACK):** Simulated a payment decline (HTTP 402) on a MacBook Pro (₦2.85M). Transaction issued an immediate `ROLLBACK`, returning stock to 10 units with **0 orphaned order records**.
- **Partial Rollback (SAVEPOINT):** Simulated digital wallet failure; transaction rolled back to `SAVEPOINT before_payment_attempt` and completed via Credit Card without re-executing cart reservations.

Multi-Threaded Concurrency Simulation (`simulate_concurrency.py`)

To prove concurrency control against the *Lost Update / Overselling Race Condition*, two parallel threads competed simultaneously for Product 8 (refrigerator, initial stock = 6) using pessimistic locking (`SELECT ... FOR UPDATE`).

```
[13:42:59.191] [SESSION 1] Connecting to database and requesting lock on product_id = 8...
[13:42:59.692] [SESSION 2] Attempting concurrent access on product_id = 8 (Expect Lock Wait)...
[13:43:02.323] [SESSION 1] Finished Execution: Stock 6 -> 5 | Transaction COMMITTED. Lock released.
[13:43:02.327] [SESSION 2] Finished Execution after 2.64s wait (Proves Lock Blocked Session 2):
                         Acquired lock, read updated stock (5), decremented to 4, and COMMITTED!
FINAL INVENTORY STATE: product_id = 8 | Verified Serial Stock = 4 units (Race Condition ELIMINATED)
```

5. PORTFOLIO ACTIVITY 4: NOSQL & ADVANCED DATA MODELS (7 MARKS)

To overcome the relational Entity-Attribute-Value (EAV) dilemma, NovaMart's catalog was modeled as rich BSON/JSON documents in MongoDB (`nosql_model.json`).

- **Polymorphic Specifications:** Products store divergent attributes seamlessly (Laptops: `ram_gb`, `processor`; Suits: `virgin_wool`, `fit_type`; Rice: `grain_length_mm`, `aging`) without schema alteration.
- **Embedded Reviews:** Customer reviews and aggregated ratings are embedded directly inside product documents, allowing product pages to be retrieved in a single $O(1)$ read without table joins.
- **Atomic Operations & Aggregation:** Verified `$push` review additions, `$inc` counter updates, and executed a 4-stage aggregation pipeline calculating category satisfaction (Electronics: 4.94★ across 6 reviews; Groceries: 4.90★ across 2 reviews; Fashion: 4.70★).
- **Redis Caching Tier:** Key-value store managing active user shopping carts with auto-expiring 7-day TTLs (`HSET cart:session:<id>`) and in-memory stock counters for flash sales.

6. PORTFOLIO ACTIVITY 5: DISTRIBUTED & CLOUD DATABASE ARCHITECTURE (6 MARKS)

NovaMart's multi-region distributed cloud topology was specified and validated through a production Docker Compose cluster (`docker-compose-distributed.yml`):

- **GTID Semi-Synchronous Replication:** A primary master node handles all write transactions, synchronizing via Global Transaction Identifiers (GTID) to dedicated read replicas with zero transaction loss ($RPO = 0$).
- **Entity-Based Horizontal Sharding:** Partitioned horizontally by `customer_id` using consistent hashing: $\text{Shard} = \text{MurmurHash3}(\text{customer_id}) \pmod N$, guaranteeing 98% of queries run as fast single-shard operations.
- **High Availability & Failover:** Multi-AZ deployment across three availability zones with automated health monitoring and Raft quorum consensus ($RTO < 30\text{ seconds}$).
- **CAP & PACELC Trade-Offs:** Enforcing **CP** (Consistency + Partition Tolerance) on financial checkout transactions, and **AP** (Availability with Eventual Consistency) on catalog browsing and reviews.

7. INDIVIDUAL TECHNICAL REFLECTION

7.1 Architectural Design Choices & Trade-Offs

The defining architectural decision was embracing **Polyglot Persistence**. While early prototypes explored storing polymorphic specifications within MySQL 8.0 `JSON` columns, querying deeply nested attributes and calculating cross-attribute facets proved computationally expensive. MongoDB's native document model eliminated the relational EAV anti-pattern, permitting zero-downtime catalog expansions. Conversely, maintaining customer financial ledgers and stock reservations in MongoDB would have sacrificed strict multi-table relational ACID guarantees. Pairing MySQL for transactional ledgers with MongoDB for product browsing achieved the optimal balance of consistency, latency, and operational flexibility.

7.2 Practical Engineering Challenges & Solutions

1. **Timezone DST Discontinuity During Bulk Seeding:** When inserting 9,000 randomized orders, MySQL threw `ERROR 1292: Incorrect datetime value` due to Daylight Saving Time (DST) spring-forward hour jumps. Enforcing explicit UTC session timestamps (`SET time_zone = '+00:00';`) resolved the issue deterministically.
2. **Eliminating Intermediate Filesorts in Analytical Queries:** In Portfolio 2, initial profiling revealed that filtering by `order_status` still incurred an expensive in-memory sort on `order_date DESC`. By aligning with B-Tree composite index mechanics—placing equality attributes before sorting attributes—the new composite index `idx_orders_status_date` traversed leaf pages in pre-sorted order, slashing latency by 42.1%.
3. **Mitigating Concurrency Lock Contention:** Pessimistic row locking (`FOR UPDATE`) proved highly effective in preventing overselling, but holding locks during slow external payment gateway calls risks thread exhaustion. In the simulation, Session 2 was blocked for 2.64s. In production, row locks must be restricted to minimal critical sections, augmented by Redis in-memory atomic decrement guards.

8. ACADEMIC INTEGRITY & MANDATORY AI-USE DECLARATION

This database portfolio represents original, independent technical work designed, developed, and verified by **Peace Ekoh (ID: 301866431)**.

Where artificial intelligence tools were utilized as learning aids in accordance with university policy:

- **AI Tool Used:** Google DeepMind / Antigravity Agentic Assistant.
- **Scope of Use:** Conceptual brainstorming of e-commerce edge-case schemas, review of MySQL 8.0 `EXPLAIN ANALYZE` syntax, and formatting markdown tables.
- **Student Verification & Validation:** Every SQL DDL script, stored procedure, transaction flow, concurrency simulation, and NoSQL aggregation pipeline was personally executed, tested, and validated on local installations of **MySQL 8.0.30**, **Python 3.10**, and **Node.js 18**. All performance metrics, execution logs, and concurrency lock timings reflect genuine empirical data generated on my local machine.

9. REFERENCES

1. Elmasri, R. and Navathe, S.B., 2021. *Fundamentals of Database Systems*. 7th ed. Boston: Pearson.
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